CHANDU PRASAD BHAIRAPU



AI Engineer (LLM) • LLM-Based Workflows & Evaluation

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ABOUT ME

AI engineer with an MSc in Artificial Intelligence and 3 years of professional experience, building and deploying AI-powered product features and LLM-based workflows since 2024. At an early-stage startup I owned an extraction pipeline end-to-end that turns complex documents into structured data, measured it against ground truth, and took precision/recall from 33%/45% to 93%/90%. I work in Python, FastAPI and React, deploy with Docker to AWS (CDK, GitLab CI/CD) and Azure, and use Claude Code as a standard part of building and debugging.

EXPERIENCE

Associate Generative AI Developer – Agentic Indexing Pipeline Engineer (Energy Sector)

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an LLM-based knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent structured data extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents; built the evaluation against an SAP-system-derived ground truth and used it as a quality check for debugging and iterating on prompts and agents, driving precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream instead of an LLM approach, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve AI-extracted knowledge-graph changes.
- Used Claude Code as standard practice with my own custom skills and agentic coding workflows, and built a Model Context Protocol (MCP) query service that gives Claude Code millisecond-latency retrieval from the knowledge graph.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, LLM APIs, MCP, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Postgres, Azure, Docker, Kubernetes**

GenAI Developer (AI Consulting Intern) – Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Built an AI-powered product feature with LangGraph that routes each request to vector database search, general inference, or summarization, and took it from prototype to deployment on AWS.
- Built the full stack: React frontend and Python backend with FastAPI, integrating the agentic workflow into the user-facing app; API testing with Postman.
- Containerized the stack with Docker and deployed it to AWS using CDK; version-controlled in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) – Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, scoring model answers against chemist-provided ground truth.
- Built AWS infrastructure with Lambda, SQS, and DynamoDB using CDK for the PrismaChat RAG evaluation.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, PyTorch, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer – Electric Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Care Note Structurer | Python | FastAPI | Open-Source LLMs | React

- Documentation and workflow automation for healthcare: turns dictated care-note transcripts into structured outputs (vitals, wound notes, measures) checked by schema validation logic, with retries and fallback between a commercial LLM API and a self-hosted open-source LLM.
- Evaluation loop with a labelled test set scoring field-level precision/recall per model and logging latency per request, run in CI; a small React review screen lets a user accept or correct each draft entry.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA

- Fine-tuned the open-source Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **AI Engineering:** LLM-based workflows, LLM APIs, prompt engineering, LangGraph, LangChain, multi-agent systems, structured data extraction, RAG, GraphRAG, MCP, evaluation of AI outputs (RAGAS, PytrecEval), Hugging Face Transformers, QLoRA fine-tuning
- **AI Development Tools:** Claude Code (custom skills, agentic coding workflows, MCP integration)
- **Programming & Fullstack:** Python, SQL, FastAPI, REST APIs, Python microservices, React, Streamlit, Postman
- **Cloud & DevOps:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3, VPC), Azure, Docker, Kubernetes, GitLab CI/CD, Git
- **Databases:** Postgres, MongoDB, Neo4j, DynamoDB, vector databases (Qdrant, Pinecone, Azure AI Search), SPARQL
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — Aug 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

CONFERENCE PUBLICATIONS

[1] Navigating the Future: An approach of Autonomous Indoor Vehicles

CERTIFICATIONS

- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)